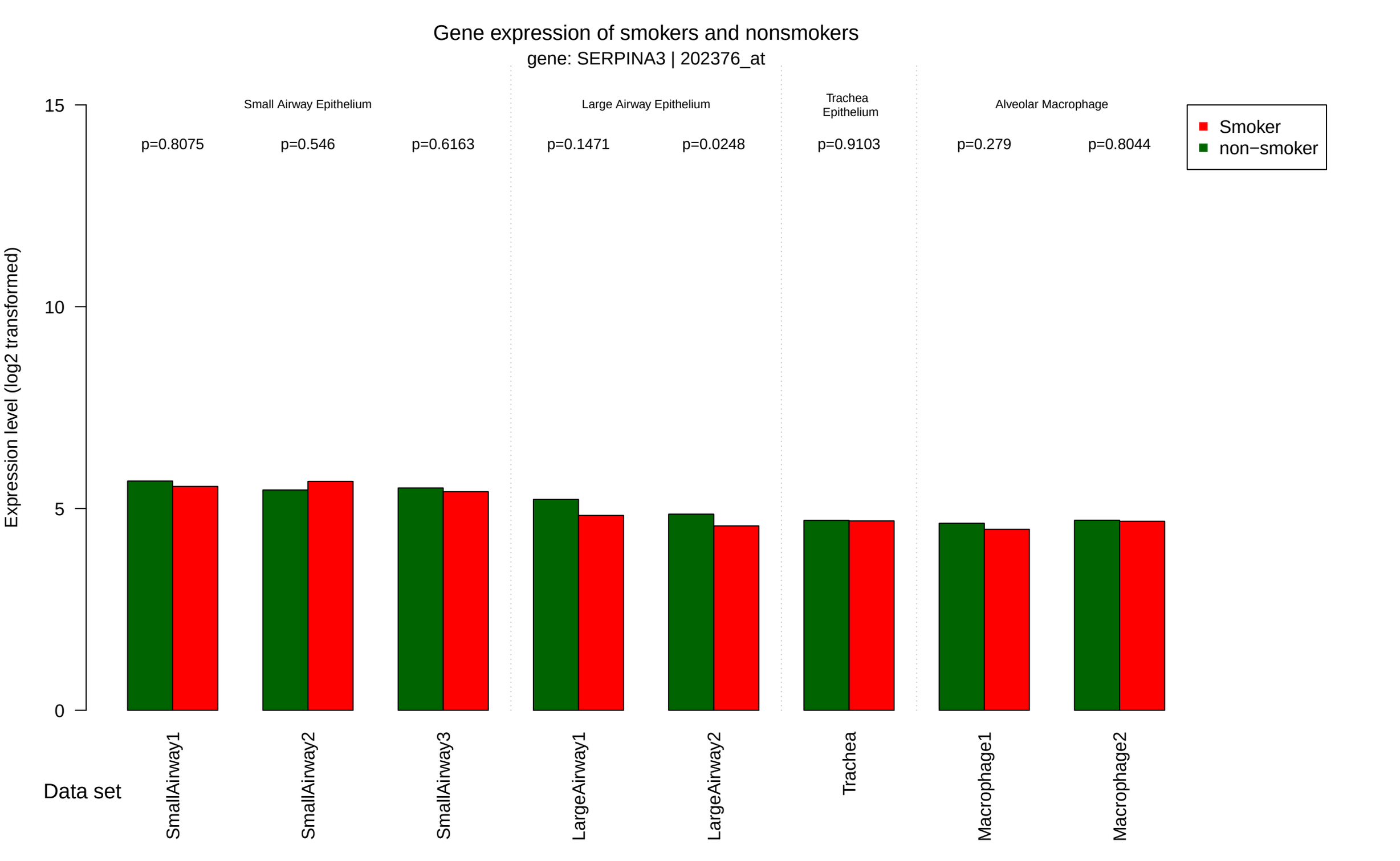
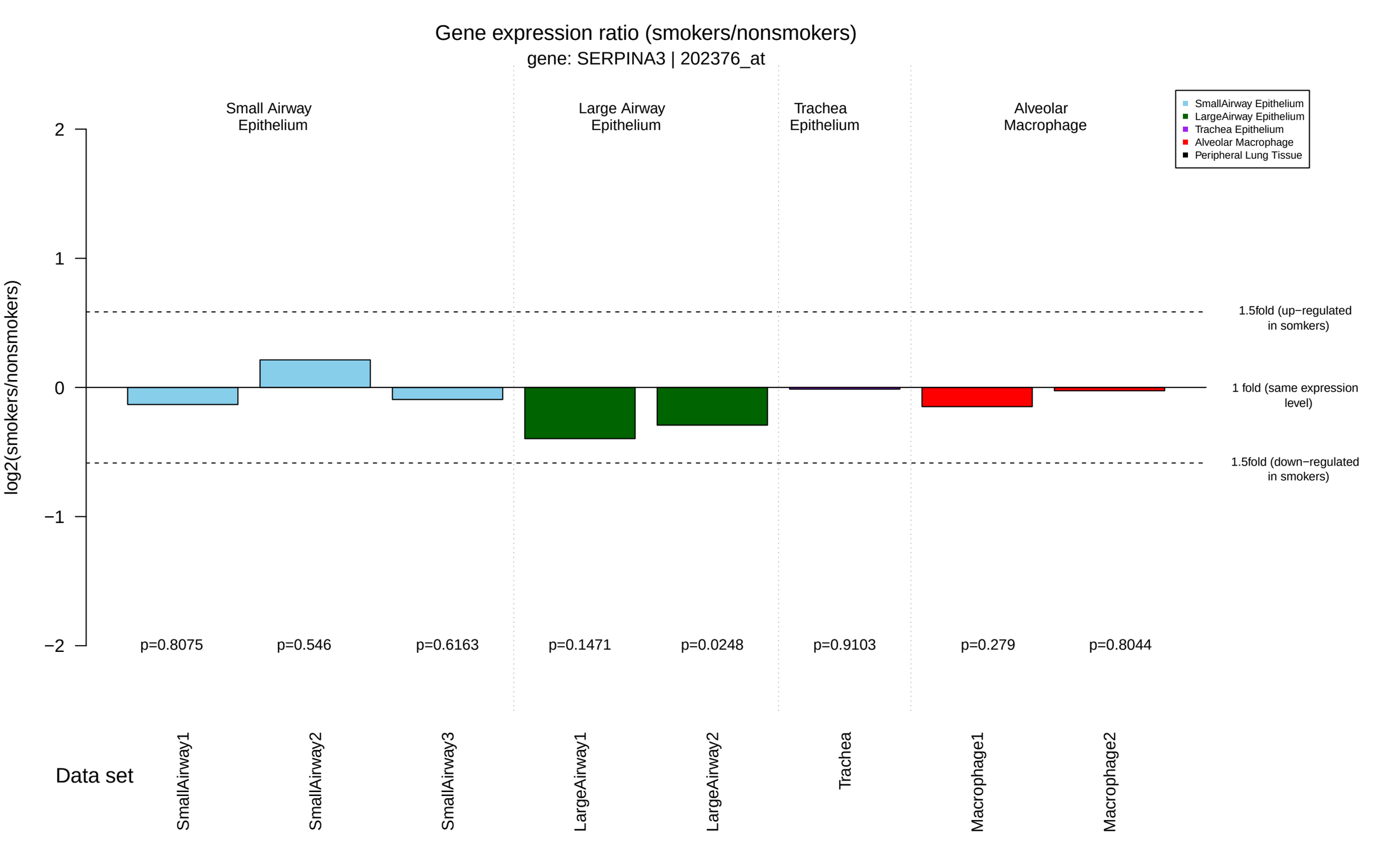
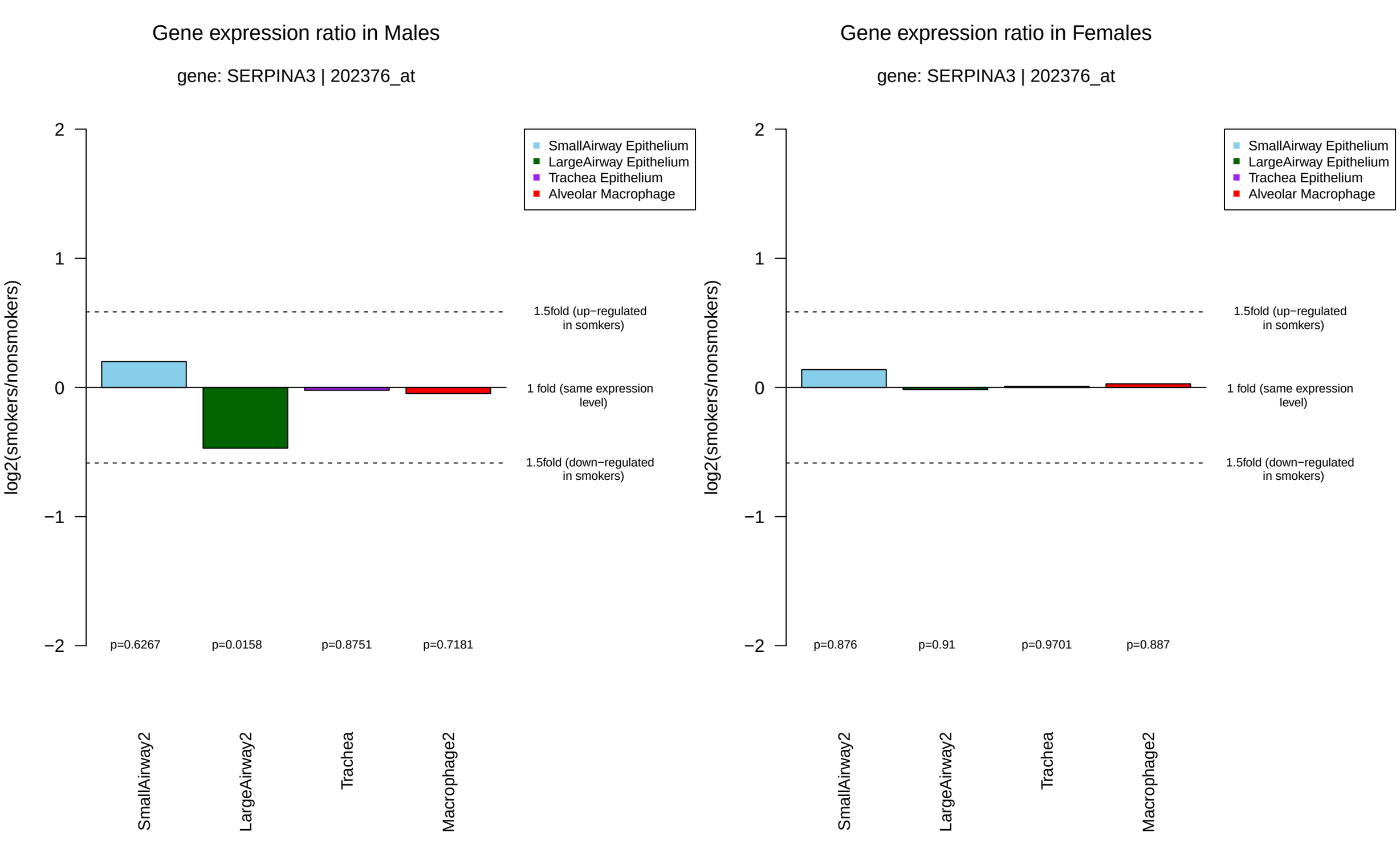
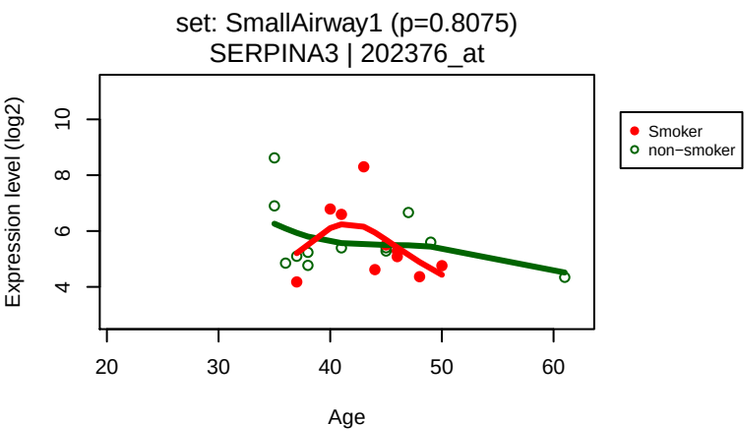
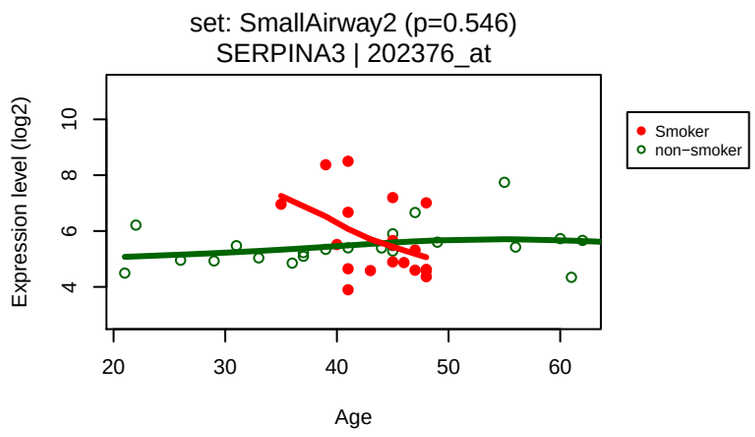
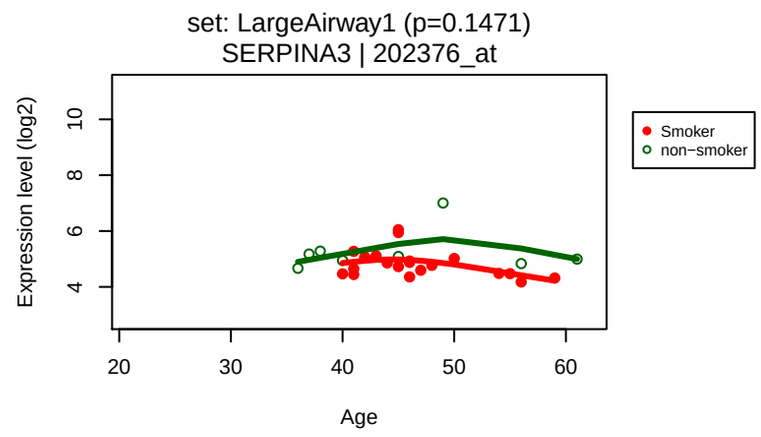

Gene expression of smokers and nonsmokers
gene: SERPINA3 | 202376_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8075
p=0.546
p=0.6163
p=0.1471
p=0.0248
p=0.9103
p=0.279
p=0.8044
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

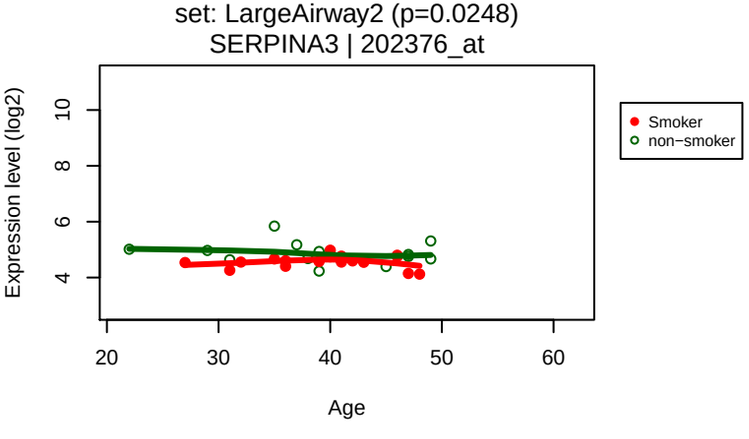

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA3 | 202376_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8075
p=0.546
p=0.6163
p=0.1471
p=0.0248
p=0.9103
p=0.279
p=0.8044
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

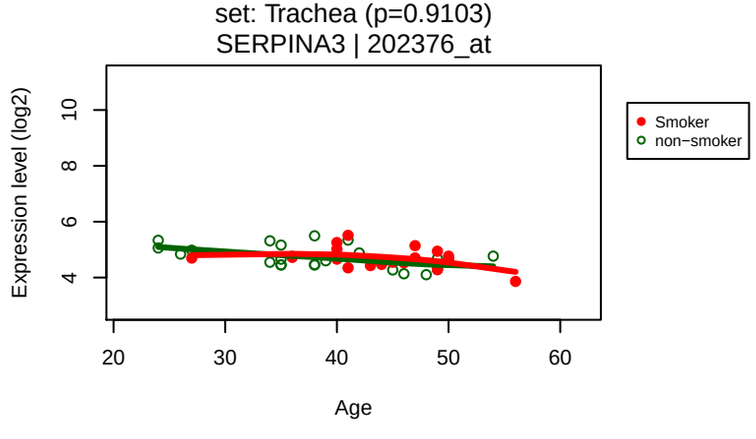

Gene expression ratio in Males
gene: SERPINA3 | 202376_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6267
p=0.0158
p=0.8751
p=0.7181
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA3 | 202376_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.876
p=0.91
p=0.9701
p=0.887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

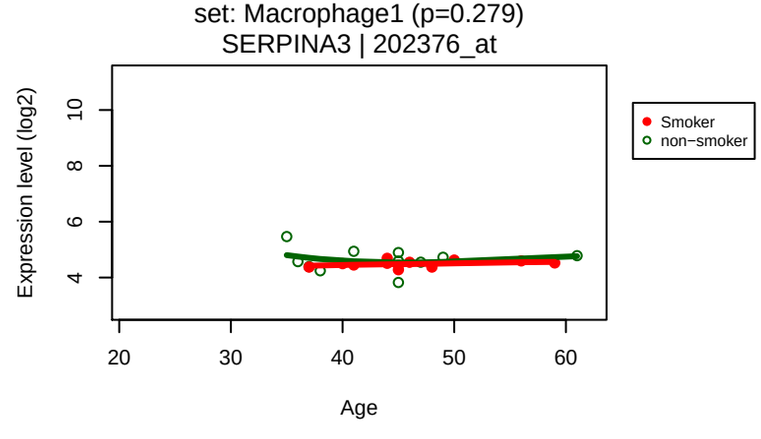

set: SmallAirway1 (p=0.8075)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker

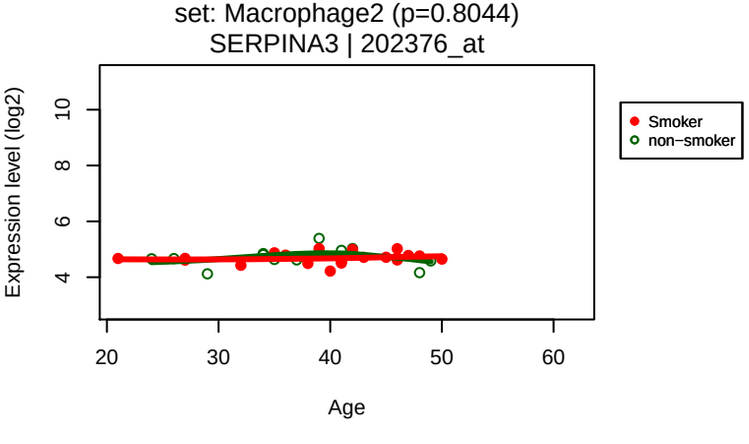

set: SmallAirway2 (p=0.546)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker

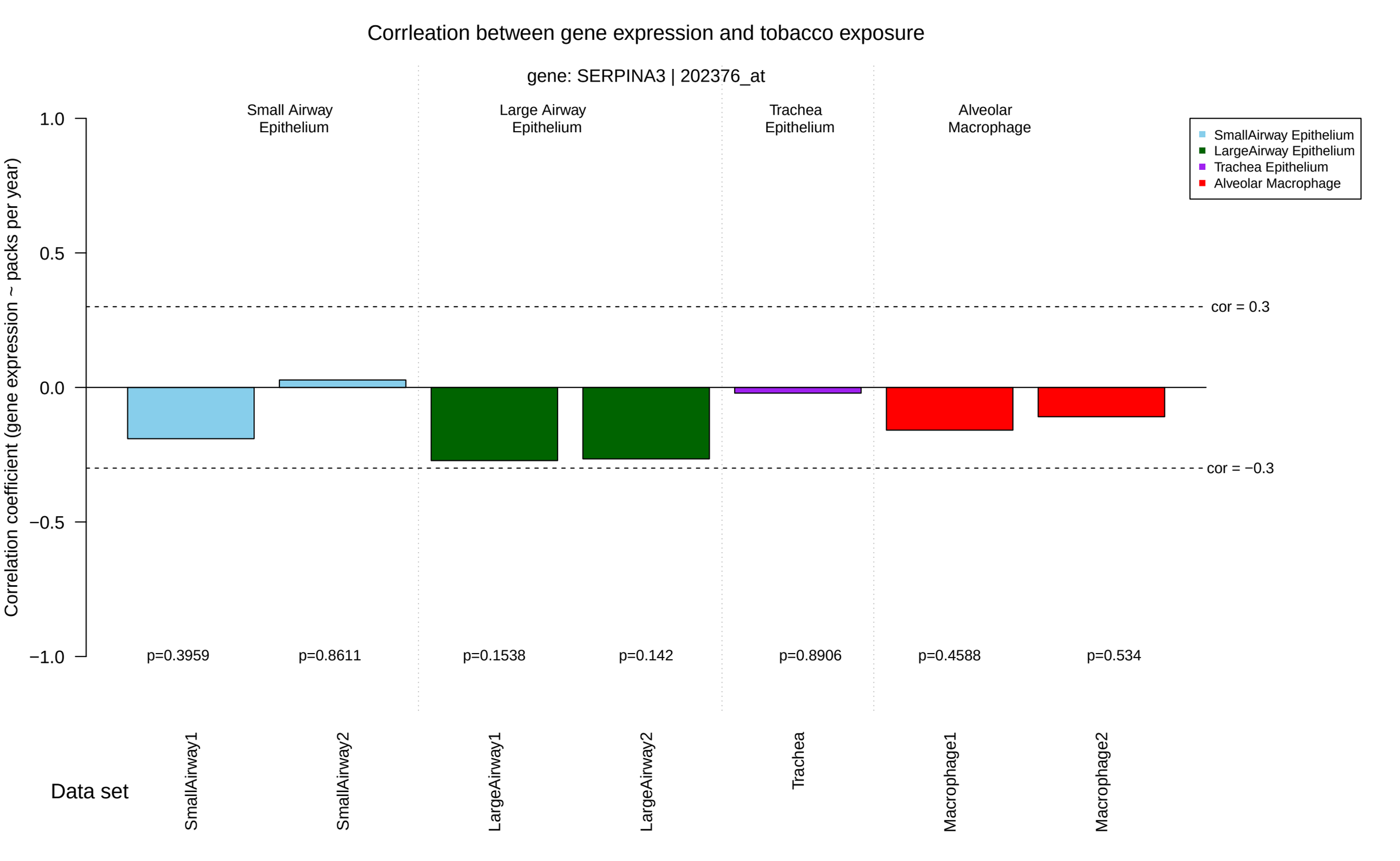

set: LargeAirway1 (p=0.1471)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0248)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9103)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.279)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8044)
SERPINA3 | 202376_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA3 | 202376_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3959
p=0.8611
p=0.1538
p=0.142
p=0.8906
p=0.4588
p=0.534
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2